

Definite integrals – 2

The mean value theorems for the integral

Homework

8 сентября 2026 года

Theoretical minimum. The first mean value theorem and the role of continuity; the second mean value theorem in both Bonnet forms; the integral analogue of Abel's summation; the mean value of a function on an interval and the intermediate value argument that produces the point ξ .

1. Prove the integral analogue of Cauchy's mean value theorem: for continuous f, g with $g > 0$ there is ξ with $\frac{\int_a^b f}{\int_a^b g} = \frac{f(\xi)}{g(\xi)}$, and explain why the naive proof by applying Cauchy's theorem to the antiderivatives needs an extra remark.
2. Let f be continuous on $[a, b]$ and let $\xi(h)$ be a point with $\int_a^{a+h} f = f(\xi(h))h$. Prove that $\frac{\xi(h)-a}{h} \rightarrow \frac{1}{2}$ as $h \rightarrow 0^+$ whenever $f'(a)$ exists and is non-zero.
3. Prove that if f is continuous on $[0, 1]$ and $\int_0^1 x^n f(x) dx = 0$ for $n = 0, 1, \dots, m$, then f has at least $m + 1$ sign changes in $(0, 1)$.
4. Let f be integrable on $[a, b]$ and put $F(x) = \int_a^x f$. Prove that for every c between $\inf f$ and $\sup f$ there are $x < y$ with $F(y) - F(x) = c(y - x)$, and decide whether x and y can always be taken arbitrarily close to each other.
5. Prove that for a non-increasing non-negative g and an integrable f one has $|\int_a^b fg| \leq g(a) \max_{t \in [a, b]} |\int_a^t f|$, and use it to estimate $\int_1^N \frac{\sin x}{x} dx$ uniformly in N .
6. Let f be continuous on $[0, 1]$. Prove that $\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) dx$ exists and identify it, first by the mean value theorem and then by an argument which does not use it.

7. Let f be continuous and let g be integrable and non-negative with $\int_a^b g = 1$. Prove that $\int_a^b fg$ lies between $\min f$ and $\max f$, and that every value in that range occurs for a suitable g .
8. Prove that if f is continuous on $[a, b]$ and $\int_a^b fg = 0$ for every continuous g with $g(a) = g(b) = 0$, then $f \equiv 0$; then weaken “continuous g ” to “polynomial g ” and prove the same.
9. Let f be continuous on $[0, 1]$ with $f(0) = 0$. Prove that there is $\xi \in (0, 1)$ with $\int_0^\xi f = \frac{\xi}{2}(f(\xi) + f(0))$ if f is in addition convex, and show that convexity cannot be dropped.
10. Prove the following sharpening of the first mean value theorem for a monotone f : the point ξ can be chosen so that $f(\xi)$ equals the mean value, and moreover ξ is unique when f is strictly monotone. Deduce a formula for the inverse function of $x \mapsto \frac{1}{x} \int_0^x f$.